

5 **(short)** A small factory consumes 20 kW of power with a lagging power factor of 0.8 at the factory. When it is connected to the 50 Hz mains power supply, the r.m.s. voltage across the factory terminals is 240 V. The line supplying the factory has an impedance $Z_s = 0.1 + j0.1 \, \Omega$.

(a) Draw a circuit diagram for the above system. Calculate the real power lost in the line. [6]

(b) Assuming that the r.m.s. voltage across the factory terminals continues to be 240 V, calculate the value of the capacitor connected across the factory terminals that is needed to correct its power factor to unity. What is the power lost in the line in this case? [4]